DSSMV Course

Desenvolvimento de Software e Sistemas Móveis (DSSMV)

Licenciatura em Engenharia de Telecomunicações e Informática

LETI/ISEP

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Disclaimer

Material and Slides

Some of the material/slides are adapted from various:

- Presentations found on the internet;
- Books;
- Web sites;
- ...

Outline

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Course

- Developing Software and Mobile Systems (DSSMV)
- Year: 2nd
- Semester: 1st
- Weeks: 15
- Classes: hours per week
  - T: 1
  - PL: 2
- European Credit Transfer and Accumulation System (ECTS) [1]: 5
  - Contact hours: **45**
    - T: 15
    - PL: 30
  - Autonomous working hours : (125 - 45 =) **80** to (140 - 45 =) **95**
    - 80 / 15 = **5.33 hours per week**.
    - 95 / 15 = **6.33 hours per week**.

[1] 1 ECTS represents 25 to 28 working hours. 5 ECTS correspond to a range between 125 and 140 working hours.

Overview

- This course focuses on mobile system topics.
- Mobile systems, like smartphones, tablets, and others, are nowadays an essential part of our lives, executing a multitude of applications, connecting us to social networks, online games, or internet calls.
  - These systems have specific characteristics, such as battery powered, many communication interfaces, and small size.
- This course aims to provide students with **skills for developing applications for mobile devices**.
- It is expected that students have acquired a good knowledge of topics taught in the courses:
  - Algoritmia e Programação (APROG);
  - Fundamentos de Desenvolvimento de Software (FSOFT).

Goals & Topic

- Goals
  - At the end of the course, the students should be able to:
    - To **explain** the generic structure of mobile operating systems
    - **Appraise** mobile operating systems.
    - **Evaluate** different approaches for developing applications for mobile systems
    - **Develop** mobile system applications
- Topics
  - The subject delivers the following main topics:
    - **Fundamental concepts** on mobile systems and underlying operating systems;
    - **Native developing** software for mobile devices;
    - **Cross-platform developing** software for mobile devices.

Teaching/learning methodology

- Lectures (T)
  - Expose students to various concepts in Mobile Computing.
- Labs (PL)
  - The learning is very **hands on** and classes are designed to allow students to practise and develop a wide range of discipline-based techniques and personal skills.
    - Students will solve a set of simple exercises.
    - Students will be encouraged to solve more complex exercises.

Public holidays

- T and PL classes are neither replaced nor recovered.

Teaching Material (I)

- Support material available at the subject's Moodle page.
  - T pdf files are used for lectures.
  - PL pdf files are used for lab classes.
  - **TP pdf files are for students' self study**.
  - **zip files are for students' self study**.
- Tools:
  - IntelliJ IDEA
  - Android Studio
  - Visual Studio Code
  - Expo

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Moments

- Assessment will have two **Moments** plus **Exam**:
  - Moment 1 (M1): Android application developing (`ProjectDroid`);
  - Moment 2 (M2): React Native application developing (`ProjectReact`).
  - Exam (E).
- These Moments (M1 and M2) and E are:
  - **Mandatory for all students, regardless of their status**.
  - M1, M2, and E are graded in the interval [0.00,20.00];

Grade

- **The DSSMV final grade (CF) is determined as follows**:
  - CF = M1 x 0.30 + M2 x 0.30 + E x 0.40.
    - CF is graded in the interval [0,20]
- **To grant access to the E**:
  - (M1 * 0,30 + M2 * 0,30) / 0,60 >= 8.00 (8.00/20,00).
- **To get E grade**
  - The minimum E grade is 8.00/20.00 (8 out of 20).

Team Work

- Team work requires **the active and equitable cooperation of all members throughout all stages of the project's development**.
- If a student demonstrates passivity, fails to attend meetings, or repeatedly fails to complete assigned tasks, **the team reserves the right to invoke the Exclusion Clause**.

Exclusion Clause

- The exclusion process must formally follow these steps:
  - **Written Warning**: The team must warn the student in writing (cc'ing professors `pbs@isep.ipp.pt` and `caf@isep.ipp.pt`) about their lack of contribution, setting a reasonable deadline for behavioral correction.
  - **Official Communication**: If the non-compliance persists, the team must formally notify professors (by email, `pbs@isep.ipp.pt` and `caf@isep.ipp.pt`) of the student's exclusion, providing evidence of the uncompleted work (e.g., version history, meeting minutes, or message logs).
  - **Deadline**: The exclusion of any member can only be validated up to two weeks before the final submission date.
    - No complaints or reports regarding a lack of group contribution will be accepted **on the day of the submission or oral defense**.
  - The excluded student will receive a grade of zero (zero) marks for the group component, unless the professor authorizes an alternative individual assignment.

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Week mapping [2]

Week	Dates	Week	Dates
1	14/09 – 20/09/2026	9	09/11 – 15/11/2026
2	21/09 – 27/09/2026	10	16/11 – 22/11/2026
3	28/09 – 04/10/2026	11	23/11 – 29/11/2026
4	05/10 – 11/10/2026	12	30/11 – 06/12/2026
5	12/10 – 18/10/2026	13	07/12 – 13/12/2026
6	19/10 – 25/10/2026	14	14/12 – 20/12/2026
7	26/10 – 01/11/2026	15	04/01 – 10/01/2027
8	02/11 – 08/11/2026	16	11/01 – 17/01/2027

- Public holidays:
  - 05/10/2026 (Implantação da República), Monday;
  - 01/12/2026 (Restauração da Independência), Tuesday;
  - 08/12/2026 (Dia da Imaculada Conceição), Tuesday;
- Christmas holiday: 21/12/2026 – 02/01/2027

[2] We consider the first week day is on Monday and the last week day is on Sunday.

Planning [3]

Week nr	Milestone
1	
2	Start `ProjectDroid` and Teamwork composition for `ProjectDroid`
3	
4	`ProjectDroid` checkpoint
5	`ProjectDroid` checkpoint
6	`ProjectDroid` checkpoint
7	Week+ISEP
8	End `ProjectDroid`
9	`ProjectDroid` presentation
10	Start `ProjectReact` and Teamwork composition for `ProjectReact`
11	`ProjectDroid` checkpoint
12	`ProjectDroid` checkpoint
13	`ProjectReact` checkpoint
14	End `ProjectReact`
15	`ProjectReact` presentation
16	

[3] Here, we consider the first week day is on **Monday** and the last week day is on **Sunday**. In this schedule, all deadlines are at **23:59** of the **Sunday** of the respective week. Therefore, whenever we mention **Week *x***, the deadline is on **Sunday** at **23:59**.

Deadline Fails

Pay Attention

For each day of delay there is a 5% penalty.

- **Repeating, hoping you will memorise**

Pay Attention

For each day of delay there is a 5% penalty.

- **Once again.**

Pay Attention

For each day of delay there is a 5% penalty.

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Authorship

- Students are **the authors of the work (materials, reports, projects, or assignments) they submit**.
  - Students are legally and educationally responsible for all work they submit for school assessment purposes
- Authorship **implies that the student has a thorough understanding of the content** and **is able to justify the choices, methods and conclusions presented**.
  - **The lack of substantiation, knowledge or a clear inability to explain** the procedures and concepts contained in the work **constitutes evidence of academic fraud** (copying, plagiarism, or undue recourse to third parties/automated tools).
- The use of Artificial Intelligence (AI) tools **is permitted to support the learning process**, provided that a statement is made regarding the tools used and their respective purposes, and that the student takes responsibility for the decisions made and the content presented.

`ProjectDroid` purpose

- The students are required:
  - To develop an Android Application project (APP) using Java programming language.
  - To write a Technical Report (TR).
  - Rules:
    - The PRJ source code must be in a repository (bitbucket).
    - The TR must be submitted electronically at `moodle.isep.ipp.pt`.
    - The TR must follow the Lecture Notes in Computer Science (LNCS) format (MS Word and Latex templates, `SurveyFormat.doc` and `SurveyFormatTex.zip`, are available at `moodle.isep.ipp.pt`).
- Teams of up to **two students**, of same PL class;

Teamwork composition

Each teamwork composition must be communicated until end of week 2 by email(`pbs@isep.ipp.pt`).

ProjectDroid Assessment (I)

ProjectDroid grade

ProjectDroid = 0.20 x TR + 0.80 x APP

- TR
  - Requirements engineering
  - Analysis & Design
  - Implementation & Tests
- APP assessment is split into three components:
  - **Checkpoint**, which refers to project development progress.
    - Graded in the interval [0.00,20.00];
  - **Implementation** (IMPL), which refers to the code quality as well as the features of the application.
    - Graded in the interval [0.00,20.00];
  - The **Factor**, that refers to the discussion with each team member (**individually**)
    - **Graded in the interval [0,100%]**;
- APP = (Checkpoint * 0.20 + IMPL * 0.80) * Factor

ProjectDroid Assessment (II)

- **Checkpoint**
  - Refers to the status of the application project (the amount of features already implemented and so on).
  - It is **mandatory an executing application project**.
- **IMPL**
  - Implementation must follow the Mobile Systems programming principles.
  - It is **mandatory a functional application** (it must run, otherwise it is not considered)
- **Factor**
  - **Factor has a great impact**
    - 0 = 20 x **0%**
    - 10 = 20 x **50%**
    - 20 = 20 x **100%**
  - Some questions were:
    - Could you explain this functionality?
    - Why are you using this approach?
    - Where is the class 'x'?

ProjectReact

- The purpose, goals and so on are similar to those defined to the `ProjectDroid` project.
- The main difference is: **To develop a Mobile Application using React Native framework**.

Teamwork composition

Each teamwork composition must be communicated until the end of week 10 by email(`pbs@isep.ipp.pt`).

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Identification

- Paulo Baltarejo Sousa.
  - E-Mail: pbs@isep.ipp.pt.
  - Office: B114
- Carlos Filipe Freitas.
  - E-Mail: caf@isep.ipp.pt.
  - Office: B117

Support (I)

- There is no specific schedule for students support.
  - Whenever students need support, must send an email to any instructor (preferably to PL instructor).



Email rule

- **Subject/Assunto**: **DSSMV**
  - If you do not follow, the email is ignored (i.e., it is not ”received”)

Support (II)

- For installing software
  - Help is provided until week 2.
  - Recall, **until week 2.**

  - Do you not understand?
  - Recall, **no help is provided after week 2.**

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Resources (Java)

- "Big Java: Early Objects", 6th Edition by Cay S. Horstmann
- "Java™:The Complete Reference", 7th Edition,Herbert Schildt
- "Java™Programming", 7th Edition, Joyce Farrell
- `https://docs.oracle.com/javase/tutorial/java/nutsandbolts/index.html`
- `http://beginnersbook.com/java-tutorial-for-beginners-with-examples/`
- `https://www.leepoint.net/index.html`
- `https://junit.org/junit5/docs/current/user-guide/`

Resources (Android)

- "Mastering Android Application Development", by Antonio Pachon Rui, 2015
- `https://developer.android.com/index.html`
- `http://simple.sourceforge.net/home.php` `http://simple.sourceforge.net/download/stream/doc/tutorial/tutorial.php`

Resources (React)

- David Flanagan, "JavaScript: The Definitive Guide", O'Reilly Media, Inc., 2020
- Adam Boduch and Roy Derks, "React and React Native: A complete hands-on guide to modern web and mobile development with React.js, 3rd Edition",Packt Publishing, 2020
- Dan Ward, "React Native Cookbook Second Edition Step-by-step recipes for solving common React Native development problems", ,Packt Publishing, 2019
- `https://reactnative.dev/`
- `https://reactjs.org/`
- `https://reactnavigation.org/`